

|                          |                                                                                                                                                |                  |                   |                    |             |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Course Code              | STA5CJ303(P)                                                                                                                                   |                  |                   |                    |             |
| Course Title             | <b>Testing of Hypothesis</b>                                                                                                                   |                  |                   |                    |             |
| Type of Course           | <b>Major</b>                                                                                                                                   |                  |                   |                    |             |
| Semester                 | V                                                                                                                                              |                  |                   |                    |             |
| Academic Level           | 300-399                                                                                                                                        |                  |                   |                    |             |
| Course Details           | Credit                                                                                                                                         | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                          | 4                                                                                                                                              | 3                | -                 | 2                  | 75          |
| Pre-requisites           |                                                                                                                                                |                  |                   |                    |             |
| Course Summary Objective | Objective make students aware of statistical hypotheses, framing of proper null and alternate hypothesis, selection of tests based conditions. |                  |                   |                    |             |

### Course Outcomes (CO):

| CO                                                                                                                                                                                             | CO Statement                                                            | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------|---------------------|--------------------------------------------------------|
| CO1                                                                                                                                                                                            | Describe the basic concepts and terminologies of testing of hypothesis. | U                | C                   | Instructor-created exams / Quiz                        |
| CO2                                                                                                                                                                                            | Understand the theory behind the statistical test construction          | Ap               | P                   | Practical Assignment / Observation of Practical Skills |
| CO3                                                                                                                                                                                            | Understand some specific statistical tests and their application        | Ap               | P                   | Seminar Presentation / Group Tutorial Work             |
| CO4                                                                                                                                                                                            | Describe the situations where parametric tests cannot be used.          | U                | C                   | Instructor-created exams / Home Assignments            |
| CO5                                                                                                                                                                                            | Understand the non-parametric alternatives of parametric tests.         | Ap               | P                   | One Minute Reflection Writing assignments              |
| CO6                                                                                                                                                                                            |                                                                         | Ap               | P                   | Viva Voce                                              |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                         |                  |                     |                                                        |

### Detailed Syllabus:

| Module     | Unit                                                 | Content                                                                                 | Hrs<br>(45<br>+30) | Marks<br>(70) |
|------------|------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------|---------------|
| <b>I</b>   | Statistical Hypothesis                               |                                                                                         | <b>9</b>           | <b>15</b>     |
|            | 1                                                    | Statistical Hypothesis definition                                                       |                    |               |
|            | 2                                                    | Null and Alternative hypothesis                                                         |                    |               |
|            | 3                                                    | Simple and Composite hypothesis                                                         |                    |               |
|            | 4                                                    | Parametric and Non parametric test                                                      |                    |               |
|            | 5                                                    | Type I and Type-II errors                                                               |                    |               |
|            | 6                                                    | Critical Region                                                                         |                    |               |
|            | 7                                                    | Level of significance & Size of the test                                                |                    |               |
|            | 8                                                    | Power of the test and p- value.                                                         |                    |               |
|            | Sections from References:                            |                                                                                         |                    |               |
| <b>II</b>  | Tests of hypothesis                                  |                                                                                         | <b>9</b>           | <b>15</b>     |
|            | 9                                                    | Most powerful test                                                                      |                    |               |
|            | 10                                                   | Uniformly Most Powerful test                                                            |                    |               |
|            | 11                                                   | Neyman- Pearson Lemma (statement and proof of sufficiency part only)                    |                    |               |
|            | 12                                                   | Application of NP Lemma to construct uniformly most powerful test,                      |                    |               |
|            | 13                                                   | Unbiased test (definition only)                                                         |                    |               |
|            | 14                                                   | Likelihood ratio test, properties of likelihood ratio tests (without proof              |                    |               |
|            | Sections from References:                            |                                                                                         |                    |               |
| <b>III</b> | Parametric Tests                                     |                                                                                         | <b>17</b>          | <b>20</b>     |
|            | 15                                                   | Large sample test concerning mean                                                       |                    |               |
|            | 16                                                   | Large sample test for equality of means,                                                |                    |               |
|            | 17                                                   | Large sample test for proportions,                                                      |                    |               |
|            | 18                                                   | Large sample test for equality of proportions.                                          |                    |               |
|            | 19                                                   | Small sample tests,                                                                     |                    |               |
|            | 20                                                   | Independent t-test, paired t-test,                                                      |                    |               |
|            | 21                                                   | Test for the significance of population variance.                                       |                    |               |
|            | 22                                                   | Concept of ANOVA                                                                        |                    |               |
|            | Sections from References:                            |                                                                                         |                    |               |
| <b>IV</b>  | Non parametric Tests                                 |                                                                                         | <b>10</b>          | <b>20</b>     |
|            | 23                                                   | Introduction and Concept                                                                |                    |               |
|            | 24                                                   | Test for randomness based on total number of runs,                                      |                    |               |
|            | 25                                                   | Empirical distribution function, One Sample Tests: Kolmogrov – Smirnov test,            |                    |               |
|            | 26                                                   | Sign test, Signed rank test (Wilcoxon)                                                  |                    |               |
|            | 27                                                   | Mann-Whitney U test. Kruskal-Wallis test.                                               |                    |               |
|            | 28                                                   | Chi-square test of goodness of fit                                                      |                    |               |
|            | 29                                                   | Chi-square test for independence of attributes                                          |                    |               |
|            | Sections from References:                            |                                                                                         |                    |               |
| <b>V</b>   | <b>Open Ended Module:<br/>Practical Applications</b> |                                                                                         | <b>30</b>          |               |
|            | 1                                                    | All statistical tests should be done to students with simple example using R or Python. |                    |               |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sections from References: |  |  |
| <p>Books and References:</p> <p>Gupta, S.C. and Kapoor, V.K. (2014): Fundamentals of Mathematical Statistics, Sultan Chand &amp; Sons.</p> <p>Christian Heumann, Michael Schomaker, Shalabh., Introduction to Statistics and Data Analysis, Springer Publications, 2016</p> <p>Goon, A.M., Gupta, M.K. and Dasgupta, B. (2002): Fundamentals of Statistics, Vol. I, 8th Edn. The World Press, Kolkata.</p> <p>Rohatgi, V. K. and Saleh, A.K. Md. E. (2009): An Introduction to Probability and Statistics. 2nd Edn. (Reprint) John Wiley and Sons</p> <p>Casella, G. and Berger R.L. (2002). : Statistical Inference, 2nd Edn. Thomson Learning</p> <p>Gibbons, J. D. and Chakraborty, S (2003): Nonparametric Statistical Inference. 4th Edition. Marcel Dekker, CRC.</p> |                           |  |  |